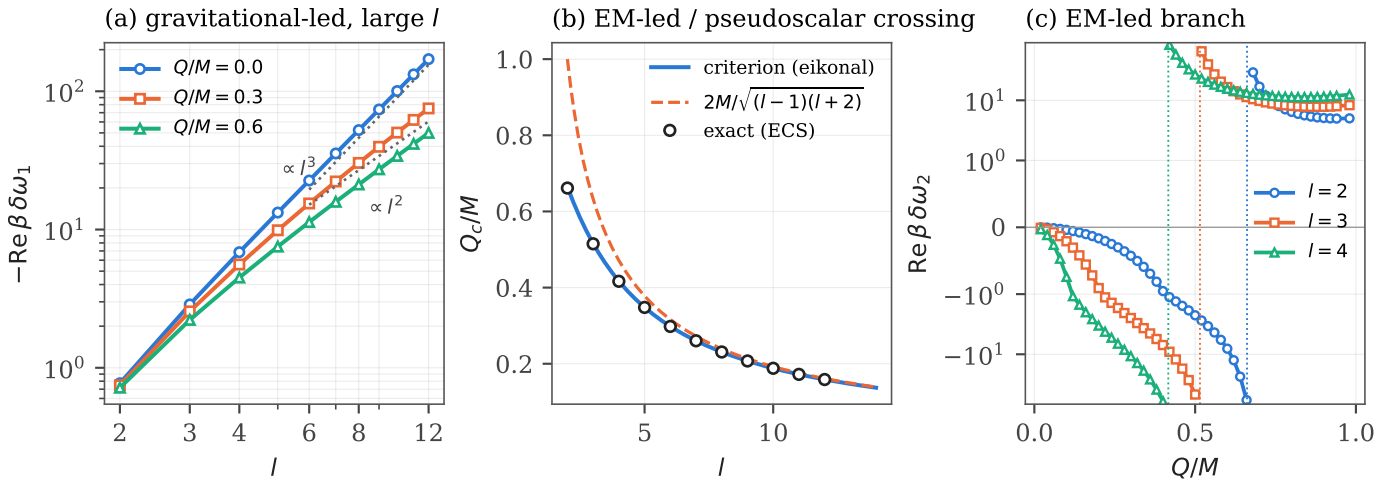
(a) gravitational-led, large l
Q/M = 0.0
Q/M = 0.3
Q/M = 0.6
∝ l^3
∝ l^2
−Re β δω1
l
(b) EM-led / pseudoscalar crossing
criterion (eikonal)
2M/√(l − 1)(l + 2)
exact (ECS)
Qc/M
l
(c) EM-led branch
l = 2
l = 3
l = 4
Re β δω2
Q/M